

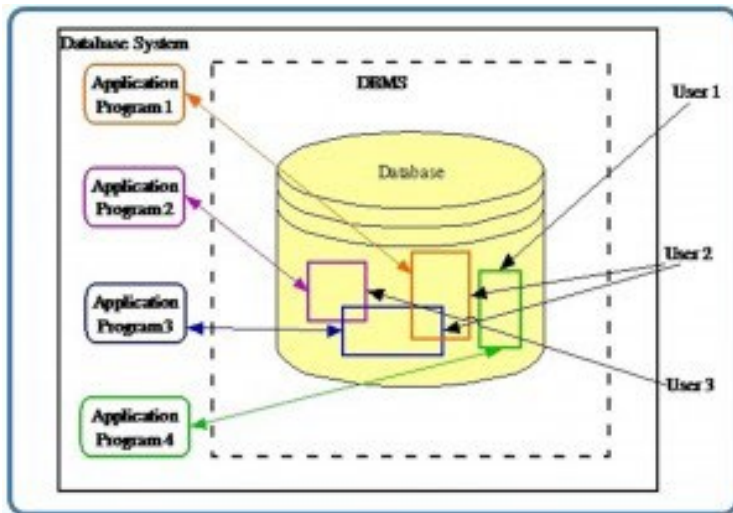
Fundamental Concepts

Database

The central idea of a database and DBMS is that instead of maintaining information in separate, isolated files, related data can be organized into a **shared database** that can be accessed by different users and applications.

What Is a Database?

A **database** is an organized, shared collection of **related data** created to support the activities of an organization.



[Relational Database](#)

This ability to organize information into **multiple related tables** is fundamental to the relational database model.

A **relational database** is a collection of related tables. In this example, the database contains the Employee and Department tables, which are related through DepartmentID.

What Is a Database Management System (DBMS)?

A database contains the **related data**.

A Database Management System (DBMS) is the **software that manages the database**.

A DBMS is a collection of programs that allows users and applications to create, maintain, retrieve, modify, and control access to a database.

Examples of DBMS software include MySQL, Microsoft SQL Server, Oracle Database, PostgreSQL, and SQLite.

Why Is a DBMS Called a "Collection of Programs"?

A DBMS performs many different tasks. It is therefore not simply one operation or command.

A DBMS includes software components responsible for activities such as:

- Creating databases and tables
- Storing data
- Retrieving data
- Updating and deleting data
- Processing queries
- Enforcing constraints
- Managing Relationships
- Controlling user access
- Handling multiple users
- Maintaining security
- Managing transactions
- Backup and recovery

When you execute:

```
SELECT *
```

```
FROM Employee;
```

you are giving an instruction to the **DBMS**. The DBMS interprets the SQL statement, determines how to retrieve the requested information, accesses the database, and returns the results.

This leads to an important distinction:

SQL is the language used to communicate with a relational DBMS. The DBMS is the software that actually manages the database.

Database vs. DBMS vs. Application Program

Consider a banking system:

An employee in the Loan

Department might use a **loan-processing application**.

The application communicates with the **DBMS**, and the DBMS accesses the **database**.

The teller does not necessarily interact directly with MySQL or write SQL commands. The **application program** can send the necessary requests to the DBMS on the user's behalf.

Different applications can therefore access the **same shared database**:

